



# KOWSALYA THIRUPATHI RAJA

## Electronics Engineer

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📍 Hamburg, Germany

### 📄 SUMMARY

Experienced in embedded development, contributing to R&D of a flow sensor at Dräger Medical technology while writing a thesis. At Bosch ebikes, enhanced test infrastructure for BMS, involving PCB design and lab systems. Conducted impactful research involving Machine Learning in EMC at TUHH. Awarded the Deutschlandstipendium and a National Research Excellence Award, driven by a passion for building reliable hardware.

### 🧠 SKILLS

**Programming & Tools:** C, Python, Verilog | KiCad, Altium | ATmega, STM32Cube IDE | LTSpice, ADS | Fusion360, Soldering | Git, VS Code | Office365, JIRA

**Others:** Research and analytical skills

### 🔗 PROFILE LINKS

LinkedIn [↗](#)  
Drive [↗](#)  
Github [↗](#)

### 🌐 LANGUAGES

**Fluent:** Tamil English Hindi  
German (B1) | Korean (A1)

### 🔑 INTERESTS

Philomath | Upskilling  
Writing | Photography

### 🛠 EXPERIENCE

#### **R&D Electronic Development - Internship and Master Thesis, DRÄGER**

01/2026 – Present | Lübeck, Germany

Developing a flow sensor based on inductance measurement and investigation of its properties. Involves Circuit Simulation, PCB designing, Soldering, Testing, Measuring, 3D modeling, associated software development, Numerical modeling of sensor behaviour, Temperature Compensation and EMC Testing.

#### **Electronic Hardware Development - Student Assistant,**

INSTITUTE OF SMART SENSORS, TUHH

09/2025 – 12/2025 | Hamburg, Germany

Embedded hardware and related software development including designing, Soldering of PCBs, Testing and Fault isolation.

#### **Research Project - Student, INSTITUTE FOR THEORETISCHE ELEKTROTECHNIK, TUHH**

10/2024 – 06/2025 | Hamburg, Germany

Involved Generating data, EDA, Building and comparing ML models for Signal integrity prediction through eye dimensions for high speed PCB interconnections.

#### **Hardware Development - Working Student,**

BOSCH EBIKE SYSTEMS, HARDWARE INFRASTRUCTURE

04/2024 – 04/2025 | Reutlingen, Germany

Requirement analysis, Component selection, PCB designing, Soldering, Validation and Testing of manufactured boards, Refining and Documentation, Test Bench calibration and more in lab support.

#### **Electronic Prototype Development - Research Assistant,**

INSTITUTE OF MICROSYSTEMS TECHNOLOGY, TUHH

07/2023 – 03/2024 | Hamburg, Germany

Involved Agile requirement adaptation, Datasheet Comprehension, Suitable Component Selection, Electrical Layout/Schematic Design, PCB Layout of Smart screw for condition monitoring in Industry 4.0.

### 🎓 EDUCATION

#### **HAMBURG UNIVERSITY OF TECHNOLOGY (TUHH),**

MSc in Microelectronics & Microsystems

10/2022 – Present | Germany

Grade: 1.6

#### **Kumaraguru College of Technology, B.E in Electronics & Communication**

08/2017 – 06/2021 | India

Grade: 1.2

### 📁 PROJECT

#### **Smart Sphere, German research foundation SPP competition, Dresden**

11/2023 – 04/2024

Localization of 3D smart sphere through the sensor data via bluetooth.